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Learning capacity allocation for stable sequential learning in recurrent spiking neural networks

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ABSTRACT

Sequential learning remains challenging due to instability and interference caused by continuous parameter adaptation. Most existing approaches focus on how parameters are updated, which implicitly assumes that adaptation is uniformly permitted across the model. Here we identify learning capacity allocation, defined as the problem of determining where adaptation is permitted, as an upstream and critical factor shaping sequential learning dynamics. We introduce IP^2-RSNN, a two-stage framework that allocates intrinsic neuronal plasticity at the task-family level and evaluates its effects during sequential learning. Across multiple cognitively motivated delayed-response task families, task-dependent intrinsic capacity allocation substantially improves learning stability and adaptation efficiency compared to misallocated intrinsic plasticity. By contrast, uniform intrinsic adaptation does not achieve the same stability as task-dependent allocation. Beyond performance, intrinsic capacity allocation induces structured specialization at both neuron and network levels, which is absent under uniform adaptation in continuous recurrent networks. These results reveal that learning capacity allocation is a principled route towards stable and interpretable sequential learning.

1. Introduction

Sequential learning over multiple related tasks poses a fundamental challenge for artificial neural systems (McCloskey & Cohen, 1989; Parisi et al., 2019), where continuous update of parameters without resetting leads to accumulation of learning dynamics, instability, and loss of previously acquired knowledge. Although this problem is often addressed through improved optimization rules or regularization techniques (Chaudhry et al., 2018; Kirkpatrick et al., 2017; Zenke et al., 2017), most existing approaches implicitly assume that adaptation is permitted over a large, pre-specified set of parameters, and focus primarily on how updates are applied, rather than determining where in the network adaptation should be permitted, a more foundational question given that learning is not free (Achille et al., 2018; Yao et al., 2020; Yoon et al., 2018) and parameter updates consume finite representational and computational resources. This implies that deciding which components of a neural system can adapt must precede or co-evolve with the design of learning rules themselves, revealing a deeper, largely overlooked problem situated upstream of update mechanisms: the strategic allocation of learning capacity across network elements for a given family of tasks. We formalize this as a problem of learning capacity allocation, deciding where adaptation is allowed in a neural system, in contrast to the common assumption of uniform adaptation (Fig. 1). Here, learning capacity means the learnable parameter subspace, that is, the parameters to which plasticity can be applied.

Recurrent spiking neural networks (RSNNs) offer a principled framework for investigating learning capacity allocation in sequential learning (Maass, 1997). As dynamical systems whose stability and computational function are exquisitely sensitive to where plasticity occurs, RSNNs naturally embody the core tension between structured adaptation and network stability (Huebotter et al., 2025; Maryada et al., 2025; Zhang et al., 2019). The intrinsic neuronal parameters of RSNNs, such as time constants and firing thresholds, can directly govern neuronal dynamics and temporal integration. Although recent studies show that these adaptive intrinsic parameters can enhance learning, existing methods typically allow them to change uniformly, without strategically allocating this form of plasticity according to task requirements (Fang et al., 2021; Shaban et al., 2021; Zheng et al., 2024). This approach overlooks

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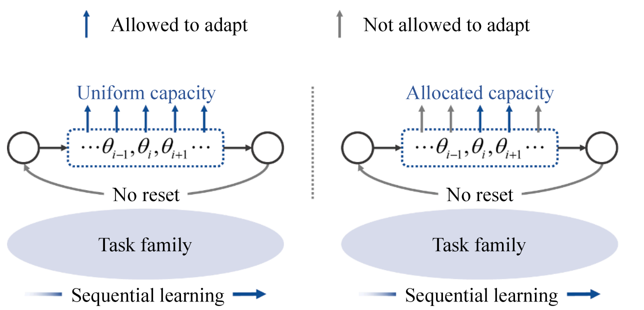

Fig. 1. A conceptual illustration of learning capacity allocation in sequential learning. Left: Uniform adaptation, where all parameters remain learnable throughout a sequence of tasks without reset. Right: Allocated adaptation, where learning is restricted to a subset of parameters shared across a task family, while the remaining parameters are fixed.

the fundamental question of capacity distribution, i.e., how a network's limited intrinsic plasticity should be optimally distributed to support stable and efficient sequential learning.

To answer this question, we introduce a two-stage framework that separates the allocation of learning capacity from its utilization. We instantiate it in a novel model, the Intrinsic Plasticity-Squared RSNN (IP^2-RSNN), which learns both intrinsic neuronal parameters and, more importantly, a policy governing where plasticity is allowed. Here, IP^2 indicates the two different roles of intrinsic plasticity in our framework. On one hand, intrinsic neuronal parameters are themselves adaptive, and on the other hand, the availability of this intrinsic plasticity is further controlled by the allocation mechanism. The framework operates in two stages: a meta-learning allocation stage that determines a structured plasticity budget for a task family, followed by an evaluation stage where sequential learning proceeds within this fixed budget. This design directly tests our core hypothesis: strategically constraining plasticity locations provides a principled mechanism to stabilize learning and mitigate interference, thereby offering a computational model to study the principles of capacity allocation.

We evaluate our framework on four cognitively motivated delayed-response task families drawn from neuroscience (Britten et al., 1992; Funahashi et al., 1989; Mante et al., 2013; Mendoza-Halliday & Martinez-Trujillo, 2017). These tasks share a common stimulus–delay–response structure but differ in their functional demands, spanning decision-based tasks that map high-dimensional stimuli to discrete actions and reproduction-based tasks that require maintaining and replaying information over time. This controlled diversity allows us to examine how learning capacity allocation varies across task classes and whether selective allocation supports stable and efficient sequential learning. Our model achieves robust performance across all task families, while substantially reducing interference and inducing structured, task-dependent specialization.

Our contributions are threefold: (1) we identify learning capacity allocation as a fundamental problem in sequential learning; (2) we introduce IP^2-RSNN, a two-stage framework for allocating intrinsic neuronal plasticity in RSNNs and evaluating its effects on sequential learning; and (3) we empirically show that task-dependent intrinsic allocations reduce interference and induce structured specialization during sequential learning.

2. Related work

2.1. Selectivity mechanisms

At the computational level, learning capacity allocation can be viewed as a specific form of selectivity. However, prior work has primarily explored selectivity in how learning or computation is organized, while leaving largely unexamined the more upstream question of where adaptation should be permitted within the system. Some meta-learning and continual learning approaches employ parameter partitioning or freezing to mitigate interference (Mallya et al., 2018; Mallya & Lazebnik, 2018; Upadhyay et al., 2025). However, in these approaches, the choice of which parameters are frozen or trainable is typically treated as a fixed design strategy, operating under an implicit allocation of learning capacity, rather than being explicitly modeled as the object of study. Another line of work introduces selective computation through routing, subspace gating, or context-dependent dynamics to enable flexible task inference (Akbarian et al., 2024; Clark & Beiran, 2025; Li et al., 2025; Liu & Wang, 2024; Zhang et al., 2025). These mechanisms primarily determine which components are engaged during inference or computation, while typically operating under the assumption that learning capacity is uniformly available and do not explicitly address where adaptation should be permitted across network components. Similarly, models incorporating intrinsic plasticity introduce selectivity at the level of neural representations by shaping selective spiking responses (Hammouamri et al., 2022; Liang et al., 2025; Tiwari & Chauhan, 2025; Zhang et al., 2021). However, in current approaches, intrinsic parameters are typically allowed to adapt uniformly whenever learning is enabled, without explicit mechanisms to allocate or restrict plasticity based on task demands. Thus, the conceptual distinction of learning capacity allocation is that it treats the location of plasticity itself as a modeling target. Instead of selecting which parameters are used for adaptation or which parameters are frozen by a predefined rule, it asks which parameter subspace should be made available for adaptation for a given family of tasks.

2.2. Sequential learning

Sequential learning, also referred to as continual learning or lifelong learning, studies how a neural system can acquire a sequence of tasks without catastrophic interference from ongoing parameter updates (McCloskey & Cohen, 1989; Parisi et al., 2019). A large body of work addresses this challenge by designing mechanisms that control how parameters are updated over time, including regularization-based consolidation methods that protect task-relevant parameters (Chaudhry et al., 2018; Kirkpatrick et al., 2017; Zenke et al., 2017), rehearsal or replay strategies that retain information from previous tasks (Bellitto et al., 2024; Rolnick et al., 2019; Wang et al., 2025), and parameter-isolation approaches that limit interference through task-specific subspaces or masks (Mallya & Lazebnik, 2018; Yoon et al., 2018). Despite their diversity, most sequential learning approaches share an implicit assumption: the set of parameters permitted to adapt is typically fixed a priori, and the primary objective is to control the magnitude, direction, or timing of updates within this set. As a result, these methods focus on mitigating interference at the level of update rules or optimization dynamics, while leaving largely unexamined a more upstream question—namely, how limited learning capacity should be allocated across components of a neural system in the first place.

2.3. Sequential learning in spiking neural networks

In spiking neural networks (SNNs), existing work has explored a range of mechanisms to mitigate catastrophic interference, often by adapting continual learning techniques such as consolidation, replay, or parameter isolation that were originally developed for artificial neural networks (ANNs) to the spiking domain (Han et al., 2023; Larionov et al., 2025; Lin et al., 2025; Skatchkovsky et al., 2022). Notably, motivated by the biological plausibility of SNNs, a subset of work draws inspiration from neural systems, incorporating mechanisms such as sparse selective activation, neuromodulatory signals, or biologically inspired consolidation strategies to regulate synaptic plasticity and reduce interference (D'Alba et al., 2025; Shen et al., 2024; Shi et al., 2025; Xiao et al., 2024). However, these approaches typically assume that learning capacity is uniformly available across SNN components and do not

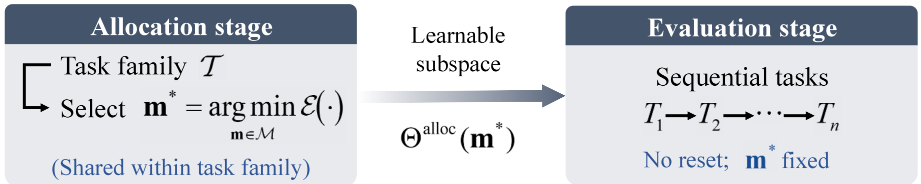

Fig. 2. Two-stage learning framework for allocating and evaluating learning capacity in sequential learning.

explicitly address how learning capacity should be allocated during sequential learning.

3. Preliminaries

We describe the neuronal dynamics of an RSNN that serves as the base model throughout this work and includes intrinsic neuronal parameters. All model parameters are treated as fixed at this stage; learning is specified in Section 4.

At each time step t, external inputs $x_{\text{in}}(t)$ and recurrent activity $S_{\text{mem}}(t-1)$ induce an effective input current

$$I(t) = \mathbf{W}_{\text{in}} x_{\text{in}}(t) + \mathbf{W}_{\text{rec}} S_{\text{mem}}(t-1) + N(t), \tag{1}$$

where $\mathbf{W}_{\text{in}}$ and $\mathbf{W}_{\text{rec}}$ denote input and recurrent synaptic weights, and $N(t)$ is additive noise. When dendritic integration is enabled (Zheng et al., 2024), inputs are first accumulated within dendritic compartments,

$$I_d(t) = \mathbf{W}_{\text{in},d} x_{\text{in}}(t) + \mathbf{W}_{\text{rec},d} S_{\text{mem}}(t-1), \tag{2}$$

and then combined to form the total input current

$$I(t) = \sum_d \left(P_{\tau_d,d} V_d(t-1) + (1 - P_{\tau_d,d}) I_d(t) \right) + N(t), \tag{3}$$

where $P_{\tau_d,d}$ and $V_d(t)$ denote the dendritic time constant and membrane potential of compartment d, respectively. When dendritic integration is disabled, the model reduces to a standard RSNN. The somatic membrane potential evolves as

$$V(t) = P_{\tau_s} V(t-1) + (1 - P_{\tau_s}) I(t), \quad S_{\text{spike}}(t) = \mathbb{1}\left[V(t) \geq P_\theta\right], \tag{4}$$

with decay factor P_{τ_s}. A spike is emitted when $V(t)$ crosses a threshold P_θ, after which the membrane potential is reset. The post-spike state evolves as

$$S_{\text{mem}}(t) = \alpha S_{\text{mem}}(t-1) + (1-\alpha) S_{\text{spike}}(t-1), \tag{5}$$

where α controls temporal smoothing. This variable represents a postsynaptic current, serving as input to recurrent and output connections.

4. Methods

We consider a sequential learning setting in which a model encounters a sequence of related tasks $T_i \in \mathcal{T}$ drawn from a common task family $\mathcal{T}$ and adapts to each task without resetting its parameters. Here, “without resetting” means that learnable model parameters, including synaptic weights and mask-enabled intrinsic parameters, are carried over across tasks. In this regime, learning updates accumulate within a shared parameter space, leading to interference across tasks. Here, we adopt an upstream perspective that focuses on learning capacity allocation, deciding where adaptation should be permitted, in order to mitigate interference at its source rather than adopting a downstream view that emphasizes how adaptation is carried out.

4.1. Two-stage learning framework

We introduce a two-stage learning framework, illustrated in Fig. 2, that explicitly allocates learning capacity by restricting plasticity to a smaller parameter subspace shared within a task family (allocation stage), and then evaluates the resulting allocation in a second stage (evaluation stage). The allocation stage of the framework does not introduce a new learning rule, but serves as a protocol for selecting and fixing an allocation prior to task-level learning. The reduced task sequence used in the allocation stage and the task stream used in the evaluation stage are generated independently from the same task-family distribution, so no task instances or task orders are shared between the two stages.

Potential learning capacity. We define the model's *potential learning capacity* as the set of all parameters

$$\Theta^{\text{pot}} = \{\theta_1, \ldots, \theta_M\}, \tag{6}$$

where M denotes the total number of parameters that could, in principle, be adapted by learning. A *capacity allocation* specifies which components of Θ^{pot} are learnable and is represented by a gating vector

$$\mathbf{m} = (m_1, \ldots, m_M) \in \mathcal{M}, \tag{7}$$

where $\mathcal{M}$ denotes the space of admissible capacity allocations and each component m_j indicates whether the corresponding parameter θ_j is learnable. The resulting allocated learning capacity is

$$\Theta^{\text{alloc}}(\mathbf{m}) = \{\theta_j \in \Theta^{\text{pot}} \mid m_j = 1\}. \tag{8}$$

Once selected, the allocation vector $\mathbf{m}$ is fixed during the subsequent evaluation stage and shared among all tasks drawn from the same task family $\mathcal{T}$.

Allocation stage. The allocation stage operates at the task-family level and selects an allocation vector $\mathbf{m}$ that induces effective learning dynamics across a sequence of tasks. Specifically, it evaluates candidate allocations based on the learning performance induced by a simulated sequential learning process that uses a reduced task sequence and limited per-task training iterations. This reduced simulation serves as an approximate probe of task-family-level learning dynamics: the performance of each candidate allocation on a short representative task stream is used to estimate whether the corresponding adaptable subspace is suitable for longer sequential learning within the same task family. Formally, the allocation stage selects

$$\mathbf{m}^* = \arg\min_{\mathbf{m}\in\mathcal{M}} \mathcal{E}\big(\mathcal{D}_{\text{sim}}(\mathbf{m}), \mathcal{T}\big), \tag{9}$$

where $\mathcal{D}_{\text{sim}}(\mathbf{m})$ denotes the simulated sequential learning dynamics under allocation $\mathbf{m}$, and $\mathcal{E}(\cdot)$ is a performance functional capturing adaptation efficiency or stability.

Evaluation stage. Given a fixed capacity allocation $\mathbf{m}$, the evaluation stage assesses the allocation by adapting only the parameters in $\Theta^{\text{alloc}}(\mathbf{m})$. For a task $T_i \in \mathcal{T}$, learning proceeds through gradient-based updates of the form

$$\theta_j^{(k+1)} = \begin{cases} \theta_j^{(k)} - \eta \nabla_{\theta_j} \mathcal{L}_{T_i}(\theta^{(k)}), & \text{if } m_j = 1, \\ \theta_j^{(k)}, & \text{if } m_j = 0, \end{cases} \tag{10}$$

where k indexes learning iterations and η denotes the learning rate. The allocation $\mathbf{m}$ thus constrains the learning dynamics, shaping both within-task adaptation and cross-task interference.

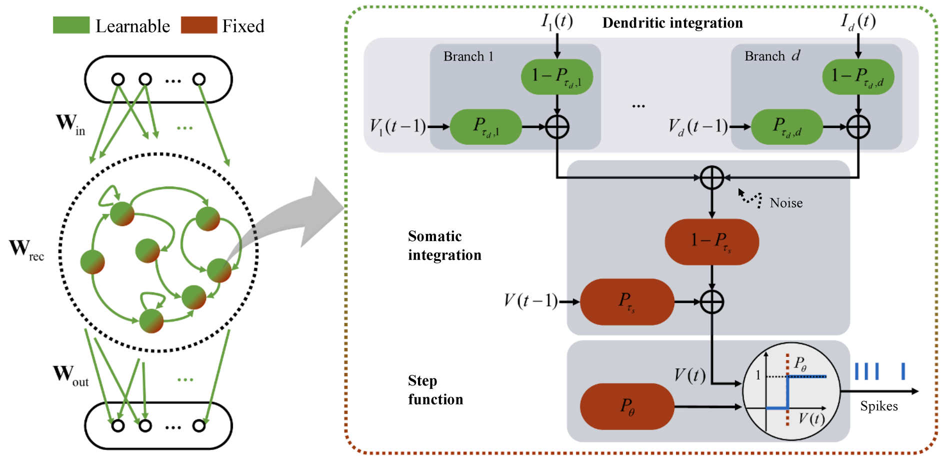

Fig. 3. Instantiation of learning capacity allocation in an intrinsic-plasticity-enabled recurrent spiking neural network (IP^2-RSNN). Here, capacity allocation operates exclusively on intrinsic neuronal parameters, whereas synaptic parameters remain fully learnable.

4.2. *IP^2-RSNN*

We now instantiate the two-stage learning framework introduced above in an intrinsic-plasticity-enabled RSNN, which we call IP^2-RSNN, as illustrated in Fig. 3 and Algorithm. 1.

Potential learning capacity of IP^2-RSNN. Building on the RSNN dynamics introduced in Section 3, IP^2-RSNN exposes intrinsic neuronal parameters as allocatable learning capacity. Formally, we instantiate the model's potential learning capacity as

$$\Theta^{\text{pot}} = \Theta^{\text{syn}} \cup \Theta^{\text{int}}, \tag{11}$$

where $\Theta^{\text{syn}} = \{\mathbf{W}_{\text{in}}, \mathbf{W}_{\text{rec}}, \mathbf{W}_{\text{out}}\}$ denotes synaptic parameters, and

$$\Theta^{\text{int}} = \{P_{\tau_d}, P_{\tau_s}, P_\theta\}, \tag{12}$$

collects intrinsic parameters governing neuronal dynamics.

Capacity allocation is applied exclusively to Θ^{int}, while synaptic parameters remain fully learnable. Restricting allocation to intrinsic parameters yields a compact and interpretable configuration space, allowing us to isolate how different task families selectively benefit from distinct forms of intrinsic plasticity.

Capacity allocation for intrinsic plasticity. Capacity allocation over intrinsic plasticity is implemented by a binary mask

$$\mathbf{m} = (m_{\tau_d}, m_{\tau_s}, m_\theta), \tag{13}$$

which gates plasticity of the intrinsic parameters. For each intrinsic parameter $P_i \in \Theta^{\text{int}}$, the effective parameter used during sequential learning is

$$\hat{P}_i = m_i\, P_i^{\text{learnable}} + (1 - m_i)\, P_i^{\text{fixed}} \quad i \in \{\tau_d, \tau_s, \theta\}. \tag{14}$$

Synaptic parameters are not gated and are equivalent to having mask values fixed to one.

As described in Section 4.1, the allocation stage selects a task-family-specific allocation $\mathbf{m}$ using simulated sequential learning on a reduced task sequence with limited per-task training iterations. The selected allocation is then fixed for the subsequent evaluation stage on that task family. For example, if an allocation $\mathbf{m}^* = (1, 0, 0)$ is selected in the allocation stage, adaptation of P_{τ_d} is allowed, while P_θ and P_{τ_s} are kept fixed during the evaluation stage.

Allocation evaluation for intrinsic plasticity. Given a fixed capacity allocation $\mathbf{m}$, the evaluation stage performs task-level learning by updating synaptic parameters and the subset of intrinsic parameters enabled by $\mathbf{m}$.

Synaptic parameters and enabled intrinsic parameters are updated via backpropagation through time. For an intrinsic parameter $\hat{P}_i$ with mask value $m_i \in \{0, 1\}$, the update rule is

$$\hat{P}_i \leftarrow \hat{P}_i - \eta\, m_i\, \nabla_{\hat{P}_i} \mathcal{L}, \tag{15}$$

so that intrinsic parameters with $m_i = 0$ remain fixed during the evaluation stage. Moreover, synaptic parameters are updated analogously without gating.

Algorithm 1: IP^2-RSNN: Two-stage intrinsic capacity allocation and evaluation.

Require: Task family $\mathcal{T}$; candidate mask set $\mathcal{M}$; RSNN with synaptic parameters Θ^{syn} and intrinsic parameters Θ^{int}; performance functional $\mathcal{E}(\cdot)$; learning rate η

Output: Selected intrinsic mask $\mathbf{m}^*$ and evaluation performance

Stage I: Allocation

for $\mathbf{m} \in \mathcal{M}$ **do**

  Initialize $(\Theta^{\text{syn}}, \Theta^{\text{int}})$;

  Simulate sequential learning on a reduced task sequence $\{T_i\}_{i=1}^{N_{\text{sim}}}$ with K_{sim} steps per task, updating

  $$\Theta^{\text{syn}} \leftarrow \Theta^{\text{syn}} - \eta \nabla_{\Theta^{\text{syn}}} \mathcal{L}, \quad P_j \leftarrow P_j - \eta\, m_j\, \nabla_{P_j} \mathcal{L},\ P_j \in \Theta^{\text{int}}$$

  Compute allocation score $s(\mathbf{m}) = \mathcal{E}\big(\mathcal{D}_{\text{sim}}(\mathbf{m}), \mathcal{T}\big)$;

Select $\mathbf{m}^* = \arg\min_{\mathbf{m} \in \mathcal{M}} s(\mathbf{m})$;

Stage II: Evaluation

Fix intrinsic plasticity mask to $\mathbf{m}^*$;

Simulate sequential learning on a full task stream $\{T_i\}_{i=1}^{N_{\text{eval}}}$ with K_{eval} steps per task, updating

$$\Theta^{\text{syn}} \leftarrow \Theta^{\text{syn}} - \eta \nabla_{\Theta^{\text{syn}}} \mathcal{L}, \quad P_j \leftarrow P_j - \eta\, m_j^*\, \nabla_{P_j} \mathcal{L},\ P_j \in \Theta^{\text{int}}$$

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5. Experiments

5.1. *Task families*

We consider four task families adapted from neuroscience experiments: delayed match-to-sample (DMS), context-dependent delayed match-to-sample (CD-DMS), and two go/no-go delayed recall families (GNG-DR-2 and GNG-DR-4) (Britten et al., 1992; Funahashi et al., 1989; Mante et al., 2013; Mendoza-Halliday & Martinez-Trujillo, 2017), see Fig. 4.

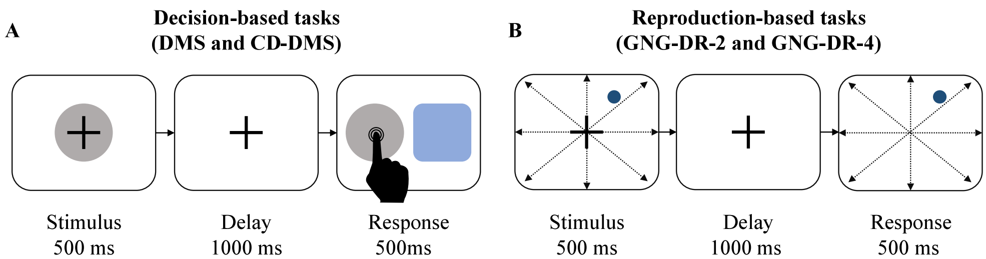

Fig. 4. Illustration of decision-based tasks (DMS and CD-DMS) and reproduction-based tasks (GNG-DR-2 and GNG-DR-4), which share a common stimulus-delay-response structure but differ in their functional demands.

Based on their underlying task rules, these task families can be grouped into two categories. Decision-based tasks (DMS and CD-DMS) require mapping high-dimensional sensory inputs to binary decisions, with CD-DMS introducing additional contextual complexity. Reproduction-based tasks (GNG-DR-2 and GNG-DR-4) require maintaining stimulus information over a delay period and reproducing it during the response phase, with GNG-DR-4 further increasing input dimensionality relative to GNG-DR-2.

The DMS family (Britten et al., 1992) follows the configuration described in Goudar et al. (2023). Each task consists of two input trials, each comprising three periods: stimulus (500 ms), delay (1000 ms), and response (500 ms). During the stimulus period, the input includes a 10-dimensional stimulus signal and a 1-dimensional fixation signal. During the delay period, only the fixation signal is maintained. During the response period, both input signals are set to zero. The corresponding target output preserves the fixation signal during the stimulus and delay periods and switches to a categorical label encoding stimulus identity during the response period.

The CD-DMS family (Mante et al., 2013) extends DMS by introducing a binary context cue during the stimulus period. When the cue is 0, the stimulus-response mapping is identical to DMS; when the cue is 1, the mapping is reversed. Therefore, each task contains four input-target trial pairs.

The GNG-DR families (Funahashi et al., 1989; Mendoza-Halliday & Martinez-Trujillo, 2017) require the model to either reproduce the input stimulus (go) or suppress the output near zero (no-go) during the response period. Two variants are considered: GNG-DR-2 with two-dimensional inputs and GNG-DR-4 with four-dimensional inputs.

5.2. Model configuration

During the allocation stage, we simulate sequential learning on a reduced task stream. For decision-based task families, we sample 20 tasks with a maximum of 20 training iterations per task. For reproduction-based task families, we sample 5 tasks with a maximum of 80 training iterations per task. The allocation performance score is computed as the average performance over the final 200 training iterations aggregated across the simulated task stream.

During the evaluation stage, we simulate sequential learning on a full task stream of 1000 tasks, with a maximum of 5000 training iterations per task. For sequential learning, the convergence criterion is set to 0.005 for the DMS, CD-DMS, and GNG-DR-2 tasks, and to 0.006 for the GNG-DR-4 task family.

Decision-based tasks are trained using cross-entropy (CE) loss, whereas reproduction-based tasks are trained using mean squared error (MSE) loss. Because these task families involve both classification and reproduction objectives, we use task-specific learning performance as the unified evaluation criterion; the number of iterations required to reach the criterion measures the adaptation efficiency. All models are trained using the Adam optimizer. The learning rate is set to 10^{-4} for the RNN and Self-Attention models, and to a higher value of 10^{-2} for all RSNN models. This choice is necessary because ANN models do not reliably converge when trained with the same learning rate configuration as RSNNs.

Table 1
Simulated sequential learning results from the allocation stage of IP^2-RSNN across four task families. The performance score $\mathcal{E}$ is reported for each intrinsic capacity mask **m**. For each task family, the best-performing mask (shown in bold) is selected as $\mathbf{m}^*$.

Mask **m**	DMS	CD-DMS	GNG-DR-2	GNG-DR-4
(0,0,0)	0.439	0.286	0.444	0.307
(0,0,1)	0.223	0.219	0.102	0.109
(0,1,0)	0.240	0.326	0.093	0.100
(1,0,0)	**0.204**	0.214	0.134	0.135
(1,1,0)	0.245	0.217	**0.074**	0.123
(1,0,1)	0.228	**0.210**	0.095	0.112
(0,1,1)	0.273	0.228	0.107	0.123
(1,1,1)	0.258	0.220	0.102	**0.079**

All RNN and RSNN models use 256 hidden neurons. For the Self-Attention models, the token dimensionality varies with the number of attention heads: 256 for single-head models, 128 for two-head models, and 64 for four-head models. When the dendritic time constant $\mathbf{P}_{\tau_d}$ is learnable, each neuron is configured with two dendritic branches and sparse connectivity, following the setup described in Zheng et al. (2024).

5.3. Task-family-specific intrinsic capacity allocation

We first examine how intrinsic learning capacity is allocated across task families during the allocation stage of IP^2-RSNN. Table 1 reports the allocation scores $\mathcal{E}$ obtained from simulated sequential learning for all candidate intrinsic masks **m** across four task families.

Different task-family-specific allocation patterns emerge, as illustrated in Fig. 5A. For decision-based tasks, the optimal mask enables adaptation of dendritic time constants only for DMS ($\mathbf{m}^* = (1, 0, 0)$), while additionally allocating plasticity to the firing threshold for CD-DMS ($\mathbf{m}^* = (1, 0, 1)$). By contrast, reproduction-based tasks consistently favor broader intrinsic plasticity: GNG-DR-2 selects adaptation of both dendritic and somatic time constants ($\mathbf{m}^* = (1, 1, 0)$), whereas GNG-DR-4 favors full intrinsic plasticity ($\mathbf{m}^* = (1, 1, 1)$). We further test the robustness of the CD-DMS allocation result under different random seeds and shuffled allocation-stage task orders. The selected mask remains $\mathbf{m} = (1, 0, 1)$, consistent with the main result; details are reported in Table B.1 in Appendix.

These results indicate that optimal intrinsic capacity allocation depends systematically on task demands, rather than favoring uniform or maximal plasticity across intrinsic parameters.

5.4. Intrinsic capacity allocation stabilizes sequential learning

Next, we evaluate how the selected intrinsic capacity allocations affect sequential learning performance. Fig. 5B compares the number of

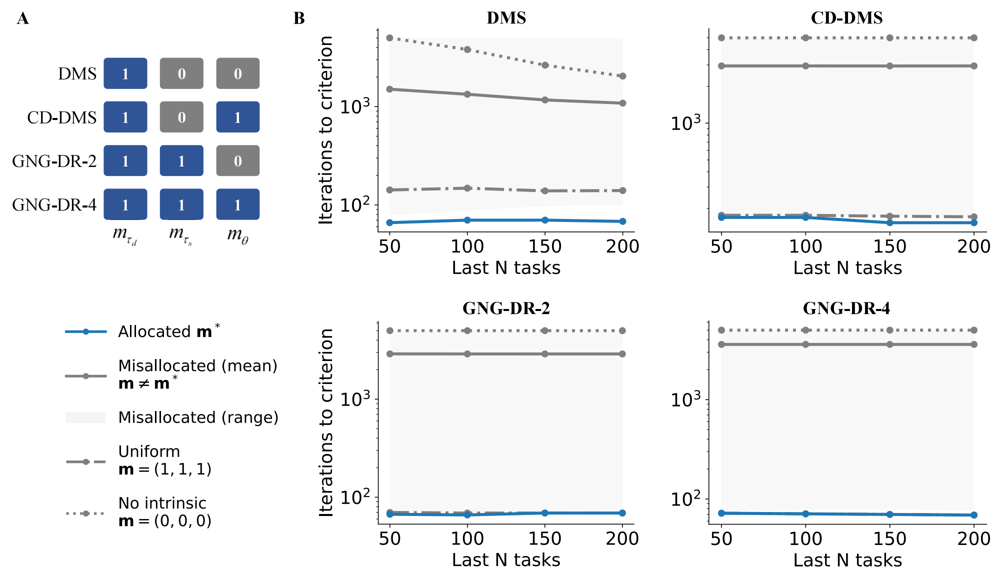

Fig. 5. Intrinsic capacity allocation and its effect on sequential learning in IP²-RSNN. (A) Task-family-specific optimal masks $\mathbf{m}^*$, where 1 denotes adaptable and 0 denotes fixed intrinsic parameters. (B) Sequential learning performance across intrinsic plasticity configurations, measured by the number of iterations required to reach the learning criterion on the final N tasks. Curves correspond to the allocated mask ($\mathbf{m}^*$), misallocated masks ($\mathbf{m} \neq \mathbf{m}^*$; mean and min-max range), uniform intrinsic plasticity ($\mathbf{m} = (1,1,1)$), and no intrinsic plasticity ($\mathbf{m} = (0,0,0)$). For GNG-DR-4, $\mathbf{m}^* = (1,1,1)$, and the allocated and uniform results overlap.

training iterations required to reach the learning criterion for the last N tasks under different intrinsic plasticity configurations.

Across all task families, the allocated configuration $\mathbf{m}^*$ yields consistently faster and more stable adaptation than misallocated masks, uniform intrinsic plasticity, or the absence of intrinsic plasticity. Misallocated configurations exhibit substantial degradation in performance, with increased training iterations. Uniform intrinsic plasticity ($\mathbf{m} = (1,1,1)$) does not reliably recover the benefits of allocation, particularly for decision-based tasks, highlighting that unrestricted intrinsic adaptation can introduce interference rather than mitigate it.

For the GNG-DR-4 task family, the optimal allocation coincides with uniform intrinsic plasticity ($\mathbf{m}^* = (1,1,1)$), and the corresponding performance curves overlap. This exception further emphasizes that allocation does not trivially favor restricted plasticity, but instead selects task-appropriate intrinsic adaptation patterns.

To examine the generality of the allocation framework, we perform two additional control experiments on the no-delay Rotated MNIST task family. For IP²-RSNN, the allocation stage selects $\mathbf{m}^* = (1,0,0)$, which also achieves the highest evaluation accuracy on independent rotation angles (Table B.2 in Appendix). For the adaptive RNN baseline with the same mask allocation mechanism and learnable time constants, the best allocation-stage mask is $\mathbf{m}^* = (1,1,0)$, which also achieves the highest evaluation accuracy (Table B.3 in Appendix).

Together, these results demonstrate that learning capacity allocation over intrinsic parameters plays a critical role in stabilizing sequential learning dynamics.

5.5. Allocation benefits extend beyond spiking architectures

To assess whether the observed instability arises from limitations specific to spiking networks, we compare IP²-RSNN against a set of ANN baselines trained under uniform adaptation, including a standard RNN and simplified Transformer-based models (Vaswani et al., 2017) with varying numbers of attention heads.

Table 2 summarizes failure counts across task families. Despite their greater representational capacity, ANN models trained with uniform adaptation exhibit frequent failures under sequential learning, particularly for CD-DMS tasks. By contrast, IP²-RSNN with allocated intrinsic plasticity achieves zero failures across all task families. Fig. 6 further illustrates this contrast using the RNN baseline. Across most task families, the RNN trained with uniform adaptation requires substantially more iterations to reach the learning criterion as sequential learning progresses, whereas IP²-RSNN maintains stable and efficient adaptation throughout the task stream.

These results indicate that instability under uniform adaptation is not specific to SNNs, but instead reflects a more general limitation of unrestricted parameter adaptation in sequential learning.

Table 2
Comparison about failure count between allocated IP²-RSNN and a set of artificial neural network (ANN) baselines trained with uniform adaptation, including a standard recurrent neural network (RNN) and three Transformer-based models (Vaswani et al., 2017) with one, two, or four attention heads (Self-Attn H1, H2, H4). Performance is measured by failure counts across task families under sequential learning, where a failure is defined as not reaching the learning criterion within the maximum number of training iterations (out of 1000 tasks). Equivalently, the task success rate is given by one minus the failure rate; zero failure therefore corresponds to a 100% task success rate.

Task family	IP²-RSNN (allocated)	ANN (uniform)			
		Self-Attn-H1	Self-Attn-H2	Self-Attn-H4	RNN
DMS	**0**	50	3	0	0
CD-DMS	**0**	1000	2	64	70
GNG-DR-2	**0**	204	0	0	0
GNG-DR-4	**0**	312	4	0	0

5.6. Task-dependent specialization induced by intrinsic capacity allocation

Finally, we investigate how intrinsic capacity allocation shapes internal network organization during sequential learning. Fig. 7A shows the evolution of dendritic time constant (τ_d) distributions across neurons over the task sequence for each task family. We observe a consistent shift in the distribution across all four task families: initially concentrated near 1, the distribution gradually becomes bimodal, with emerging peaks near 0 and 1.

To assess the functional relevance of the observed τ_d specialization, we perform targeted damage experiments by selectively silencing neuron groups stratified by their τ_d values. Fig. 7B shows the number of

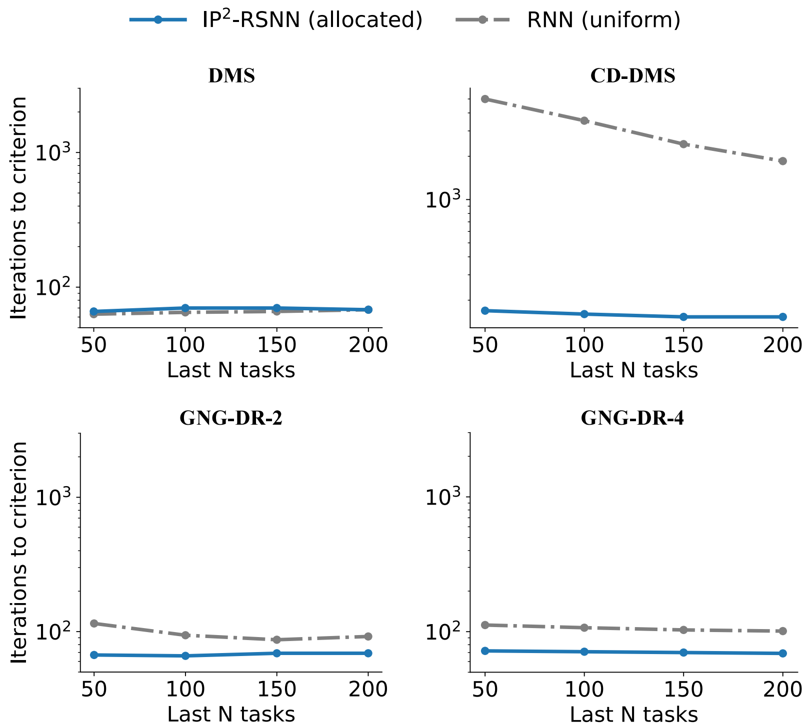

Fig. 6. Comparison between allocated IP²-RSNN and a continuous recurrent neural network (RNN) trained with uniform adaptation. Performance is measured by the number of iterations required to reach the learning criterion for the last N tasks.

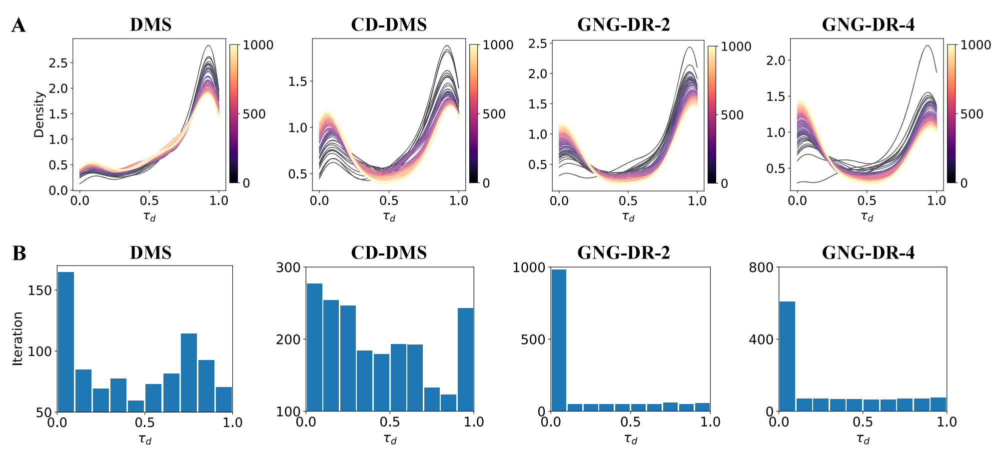

Fig. 7. Task-dependent neuron-level specialization induced by intrinsic capacity allocation. (A) Distributions of dendritic time constants (τ_d) across neurons during sequential learning, shown separately for each task family. Colors denote task indices. (B) Effects of targeted neuronal silencing based on τ_d groupings, quantified by the number of iterations required to reach the learning criterion on the subsequent task.

iterations required to reach the learning criterion on the subsequent task after silencing neurons within different τ_d ranges. For reproduction-based tasks (GNG-DR-2 and GNG-DR-4), the damage results reveal a highly asymmetric pattern: silencing neurons with small τ_d values (0-0.1) leads to a pronounced increase in adaptation iterations, whereas silencing neurons in other τ_d ranges has little effect. This indicates that a compact subset of neurons dominates task adaptation, while the remaining neurons are largely redundant. By contrast, decision-based tasks (DMS and CD-DMS) exhibit a more distributed pattern, in which silencing neurons across a broader range of τ_d values affects adaptation to varying degrees.

These results suggest that instead of requiring fast or slow neurons for each task family, intrinsic capacity allocation induces task-dependent temporal specialization. Small τ_d values correspond to fast-tracking dendritic dynamics, whereas large τ_d values correspond to slower temporal integration. The damage experiments indicate that reproduction-based tasks rely disproportionately on a compact group of fast dendritic neurons, while decision-based tasks recruit neurons across a broader range of dendritic time constants. Thus, different task families appear to use different mixtures of temporal regimes, rather than a single fast-or-slow strategy.

We further examine whether this specialization is reflected at the network level. In IP²-RSNN, we find that for both decision-based tasks (DMS and CD-DMS) and reproduction-based tasks (GNG-DR-2 and GNG-DR-4), the relationship between network metrics (modularity, community number, and stationarity, see Appendix A) and adaptation performance is consistent within each task category (Fig. 8). By contrast, in an RNN trained with uniform adaptation, no such consistent relationship is observed across tasks or metrics (Fig. 9), indicating the absence of task-dependent network-level specialization.

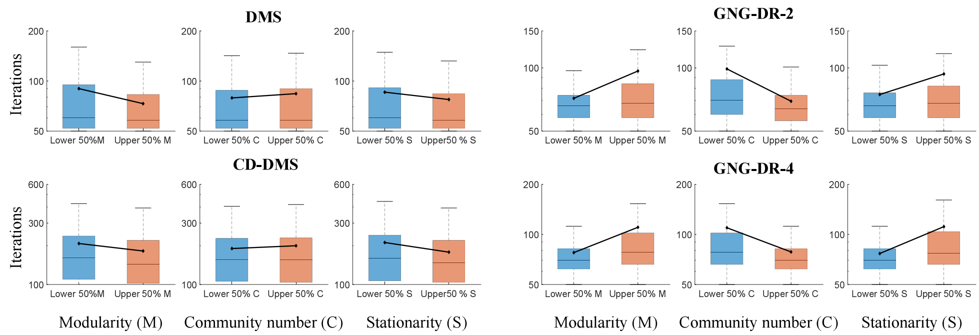

Fig. 8. Task-dependent network-level specialization induced by intrinsic capacity allocation in IP^2-RSNN. The connecting lines are provided solely as visual guides to facilitate comparison between the corresponding bins in terms of mean iteration counts.

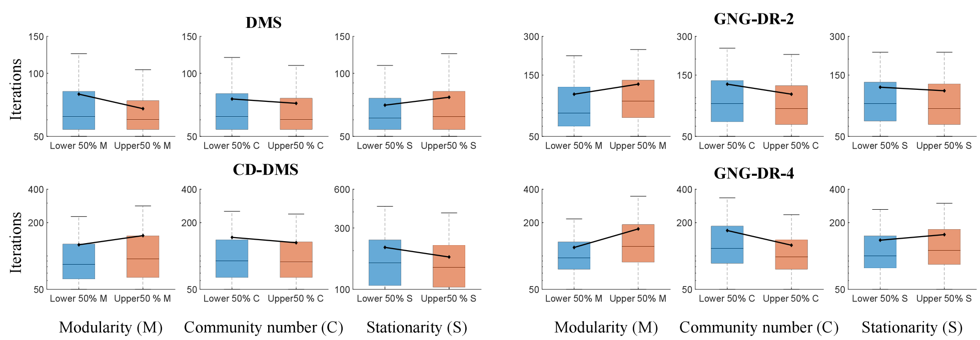

Fig. 9. Absence of task-dependent network-level specialization under uniform adaptation in an RNN. Connecting lines are included solely as visual guides, consistent with Fig. 8.

These findings demonstrate that intrinsic capacity allocation induces meaningful neuron- and network-level specialization that supports stable sequential learning.

6. Discussion and conclusion

Sequential learning over families of related tasks poses a fundamental challenge for neural systems, as unconstrained parameter adaptation can lead to instability and interference. In this work, we identify learning capacity allocation, defined as the problem of deciding where adaptation is permitted, as an upstream and largely overlooked factor shaping sequential learning dynamics.

We introduce IP^2-RSNN, a two-stage framework that explicitly allocates intrinsic neuronal plasticity at the task-family level and evaluates its effects during sequential learning. Across multiple cognitively motivated task families, we show that task-dependent intrinsic capacity allocation substantially improves stability and adaptation efficiency compared to uniform or misallocated intrinsic plasticity. Importantly, uniform intrinsic adaptation does not reliably recover the benefits of allocation, indicating that unrestricted plasticity can itself introduce interference.

Beyond behavioral performance, we demonstrate that intrinsic capacity allocation induces structured specialization at multiple levels. At the neuron level, intrinsic parameters evolve toward task-dependent regimes associated with distinct functional roles. At the network level, consistent relationships emerge between structural organization and adaptation performance within task families. Such relationships are absent in continuous recurrent networks trained with uniform adaptation. Together, these results indicate that capacity allocation not only stabilizes learning, but also organizes internal representations in a task-appropriate manner. A possible mechanism underlying these effects is that restricting plasticity reduces the number of directions in which task-induced updates can perturb the recurrent dynamics. Uniform intrinsic adaptation updates all intrinsic parameters, which provides greater flexibility but may also interfere with dynamic structures acquired from previous tasks that could otherwise benefit the new task. By contrast, task-dependent capacity allocation confines intrinsic adaptation to a selected subspace, preserving the remaining intrinsic dynamics as a stable scaffold. This balance between restricted adaptability and preserved dynamics may reduce cross-task interference while still allowing efficient adaptation. The observed neuron- and network-level specialization is consistent with this interpretation: allocated plasticity promotes task-appropriate temporal regimes and network organization, which in turn supported more stable sequential learning.

More broadly, our findings suggest that deciding where learning occurs is as critical as deciding how learning is performed. Learning capacity allocation provides a complementary perspective to existing continual learning approaches and offers a principled route toward stable, interpretable, and efficient sequential learning in both spiking and artificial neural systems. From a broader perspective, our finding that task-dependent intrinsic parameters organize fast and slow temporal regimes is related to studies that emphasize multi-timescale mechanisms for memory integration, such as astrocyte-neuron coupling dynamics for coordinating short- and long-term memory (Yin et al., 2026).

Future work will explore more scalable and adaptive mechanisms for learning capacity allocation. While the present framework performs allocation at the task-family level using a reduced sequential learning simulation, extending it to online and signal-driven schemes, such as gradient-based or activity-modulated plasticity gating, may enable adaptive allocation without explicit enumeration of candidate subspaces. Beyond intrinsic neuronal parameters, extending capacity allocation to synaptic weights represents a promising direction for further improving stability and efficiency in sequential learning. Finally, evaluating the framework across a broader range of task families and benchmark datasets will be important for assessing its generality and applicability in more complex continual learning settings.

CRediT authorship contribution statement

Yingchao Yu: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis; **Yaochu Jin:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization; **Kuangrong Hao:** Writing – review & editing; **Yuchen Xiao:** Writing – review & editing; **Yuping Yan:** Writing – review & editing; **Hengjie Yu:** Writing – review & editing; **Zeqi Zheng:** Writing – review & editing; **Wenxuan Pan:** Writing – review & editing.

Declaration of competing interest

The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data availability

The data and code supporting the findings of this study will be made publicly available on GitHub upon acceptance of the manuscript.

Appendix A. Structural modularity analysis

Structural modularity analysis involves decomposing a system into subsystems based on structural relationships. We assess modular organization using three key metrics: modularity, number of communities, and modular stationarity, and follow the configuration described in work (Gu et al., 2024).

Modularity Q quantifies the modular organization of the functional connectivity network and is defined as:

$$Q = \frac{1}{2\mu} \sum_{ijlr} \left[F_{ijl} - \gamma_l \frac{k_{il} k_{jl}}{2m_l} \delta_{lr} + \delta_{ij} C_{jlr} \right] \delta(g_{il}, g_{jr}), \tag{A.1}$$

here, i and j index neurons, l and r denote temporal layers (sliding windows over hidden-layer activity). F_{ijl} is the adjacency (e.g., Pearson correlation) between nodes i and j in layer l. γ_l controls resolution. $k_{il} = \sum_j F_{ijl}$ is the degree of node i in layer l, and $m_l = \frac{1}{2} \sum_{ij} F_{ijl}$ is the total edge weight. C_{jlr} denotes the inter-layer coupling strength linking node j across layers l and r, encouraging temporal continuity in community assignments. δ_{lr} and δ_{ij} are Kronecker delta functions, and g_{il} denotes the community label of node i in layer l. The modularity Q is maximized to identify optimal communities across all time layers.

Community number quantifies the granularity of modular decomposition, with larger values indicating more fragmented network structures. For temporal layer l, the community number C_l is defined as the number of distinct community labels assigned across all nodes:

$$C_l = \left| \{ g_{il} \mid i = 1, \dots, n \} \right|, \tag{A.2a}$$

where $| \cdot |$ denotes the cardinality of the set, and n is the total number of nodes. The overall community number $\bar{C}$ is then defined as the average across all L temporal layers:

$$\bar{C} = \frac{1}{L} \sum_{l=1}^{L} C_l. \tag{A.2b}$$

Stationarity quantifies the temporal consistency of community structures. For a given community c, its stationarity ζ_c is defined as:

$$\zeta_c = \frac{1}{L_c - 1} \sum_{l=1}^{L_c - 1} \mathrm{corr}(\mathbf{v}_{c,l}, \mathbf{v}_{c,l+1}), \tag{A.3a}$$

where $\mathbf{v}_{c,l}$ is the membership indicator vector for community c at time window l, and L_c is the number of time windows in which community c is detected. The overall stationarity $\bar{S}$ is computed as the average across all detected communities C:

$$\bar{S} = \frac{1}{|C|} \sum_{c \in C} \zeta_c. \tag{A.3b}$$

Higher values of $\bar{S}$ indicate more stable modular structures over time.

Appendix B. Additional experiments

See (Tables B.1–B.3)

Table B.1
Robustness of the allocation performance on the CD-DMS task family across random seeds and the task order. For both the fixed and shuffled task orders, three random seeds (42, 43, 44) are used. The score is computed as the mean loss over the final 200 training iterations for each seed and reported as mean ± standard deviation across seeds. Lower values indicate better performance. The mask $(1, 0, 1)$ remains the best-performing allocation pattern in both settings.

Mask **m**	Fixed task order	Shuffled task order
(0,0,0)	0.406 ± 0.170	0.702 ± 0.469
(0,0,1)	0.314 ± 0.002	0.724 ± 0.579
(0,1,0)	0.410 ± 0.189	0.601 ± 0.104
(1,0,0)	0.305 ± 0.067	0.343 ± 0.099
(1,1,0)	0.340 ± 0.109	0.305 ± 0.042
(1,0,1)	$\mathbf{0.292 \pm 0.040}$	$\mathbf{0.289 \pm 0.030}$
(0,1,1)	0.553 ± 0.097	0.466 ± 0.112
(1,1,1)	0.370 ± 0.116	0.331 ± 0.064

Table B.2
Allocation and evaluation performance on the no-delay Rotated MNIST task family. Each task is a 10-way digit classification problem under a specific image rotation angle. The rotated image is presented to the recurrent network as a short 4-step input sequence, but the task does not contain a stimulus-delay-response structure. The allocation stage uses rotation angles $0°, 30°, 60°, 90°, 120°$, while the evaluation stage uses independent angles $15°, 45°, 75°, 105°, 135°$. For each mask **m**, accuracy is averaged over angles for each seed and reported as mean ± standard deviation across three random seeds. Higher accuracy indicates better performance. The mask $(1, 0, 0)$ achieves the best performance in both allocation and evaluation stages.

Mask m	Allocation acc.	Evaluation acc.
(0,0,0)	0.944 ± 0.001	0.965 ± 0.001
(0,0,1)	0.943 ± 0.001	0.964 ± 0.000
(0,1,0)	0.946 ± 0.000	0.965 ± 0.001
(1,0,0)	$\mathbf{0.947 \pm 0.001}$	$\mathbf{0.966 \pm 0.001}$
(1,1,0)	0.946 ± 0.000	0.965 ± 0.000
(1,0,1)	0.945 ± 0.000	0.964 ± 0.000
(0,1,1)	0.945 ± 0.000	0.965 ± 0.001
(1,1,1)	0.945 ± 0.001	0.964 ± 0.001

Table B.3
Allocation and evaluation performance of the adaptive RNN on the no-delay Rotated MNIST task family. Each task is a 10-way digit classification problem under a specific image rotation angle. The rotated image is presented to the recurrent network as a short 4-step input sequence, but the task does not contain a stimulus-delay-response structure. The adaptive RNN preserves the same recurrent architecture, adaptive variables, and mask allocation mechanism as the RSNN; the only modification is the replacement of the spike-based neuronal activation with a continuous ReLU activation. The allocation stage uses rotation angles $0^\circ, 30^\circ, 60^\circ, 90^\circ, 120^\circ$, while the evaluation stage uses independent angles $15^\circ, 45^\circ, 75^\circ, 105^\circ, 135^\circ$. For each mask $\mathbf{m}$, accuracy is averaged over angles for each seed and reported as mean ± standard deviation across three random seeds. Higher accuracy indicates better performance. The mask $(1, 1, 0)$ achieves the best performance in both allocation and evaluation stages.

Mask m	Allocation acc.	Evaluation acc.
(0,0,0)	0.937 ± 0.004	0.928 ± 0.010
(0,0,1)	0.937 ± 0.006	0.920 ± 0.003
(0,1,0)	0.942 ± 0.004	0.943 ± 0.003
(1,0,0)	0.943 ± 0.005	0.942 ± 0.007
(1,1,0)	$\mathbf{0.943 \pm 0.003}$	$\mathbf{0.944 \pm 0.003}$
(1,0,1)	0.941 ± 0.002	0.938 ± 0.011
(0,1,1)	0.940 ± 0.004	0.939 ± 0.001
(1,1,1)	0.939 ± 0.002	0.941 ± 0.004

References

Achille, A., Eccles, T., Matthey, L., Burgess, C., Watters, N., Lerchner, A., & Higgins, I. (2018). Life-long disentangled representation learning with cross-domain latent homologies. *Advances in Neural Information Processing Systems*, *31*.

Akbarian, P., Nguyen, H., Han, X., & Ho, N. (2024). Quadratic gating functions in mixture of experts: A statistical insight. arXiv:2410.11222.

Bellitto, G., Proietto Salanitri, F., Pennisi, M., Boschini, M., Bonicelli, L., Porrello, A., Calderara, S., Palazzo, S., & Spampinato, C. (2024). Saliency-driven experience replay for continual learning. *Advances in Neural Information Processing Systems*, *37*, 103356–103383.

Britten, K. H., Shadlen, M. N., Newsome, W. T., & Movshon, J. A. (1992). The analysis of visual motion: A comparison of neuronal and psychophysical performance. *Journal of Neuroscience*, *12*(12), 4745–4765.

Chaudhry, A., Dokania, P. K., Ajanthan, T., & Torr, P. H. S. (2018). Riemannian walk for incremental learning: Understanding forgetting and intransigence. In *Proceedings of the european conference on computer vision* (pp. 532–547).

Clark, D. G., & Beiran, M. (2025). Structure of activity in multiregion recurrent neural networks. *Proceedings of the National Academy of Sciences*, *122*(10), e2404039122.

D'Alba, F., Kushawaha, N., Fruzzetti, L., Paolucci, P. S., & Falotico, E. (2025). Semantization of memories in a hippocampal-cortical spiking neural network. *Neurocomputing*, *640*, 130323.

Fang, W., Yu, Z., Chen, Y., Masquelier, T., Huang, T., & Tian, Y. (2021). Incorporating learnable membrane time constant to enhance learning of spiking neural networks. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 2661–2671).

Funahashi, S., Bruce, C. J., & Goldman-Rakic, P. S. (1989). Mnemonic coding of visual space in the monkey's dorsolateral prefrontal cortex. *Journal of Neurophysiology*, *61*(2), 331–349.

Goudar, V., Peysakhovich, B., Freedman, D. J., Buffalo, E. A., & Wang, X.-J. (2023). Schema formation in a neural population subspace underlies learning-to-learn in flexible sensorimotor problem-solving. *Nature Neuroscience*, *26*(5), 879–890.

Gu, S., Mattar, M. G., Tang, H., & Pan, G. (2024). Emergence and reconfiguration of modular structure for artificial neural networks during continual familiarity detection. *Science Advances*, *10*(30), eadm8430.

Hammouamri, I., Masquelier, T., & Wilson, D. (2022). Mitigating catastrophic forgetting in spiking neural networks through threshold modulation. *Transactions on Machine Learning Research*.

Han, B., Zhao, F., Zeng, Y., Pan, W., & Shen, G. (2023). Enhancing efficient continual learning with dynamic structure development of spiking neural networks. arXiv:2308.04749.

Huebotter, J., Lanillos, P., van Gerven, M., & Thill, S. (2025). Spiking neural networks for continuous control via end-to-end model-based learning. arXiv:2509.05356.

Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A. et al. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, *114*(13), 3521–3526.

Larionov, D., Bazenkov, N., & Kiselev, M. (2025). Continual learning with columnar spiking neural networks. *Optical Memory and Neural Networks*, *34*(Suppl 1), S58–S71.

Li, J., Wang, B., Zhou, X., & Hu, X. (2025). Dynamic expert specialization: Towards catastrophic forgetting-free multi-domain MoE adaptation. In *Proceedings of the 2025 conference on empirical methods in natural language processing* (pp. 18489–18504).

Liang, X., Wu, Q., Liu, W., Zhou, Y., Tan, C., Yin, H., & Sun, C. (2025). Intrinsic plasticity coding improved spiking actor network for reinforcement learning. *Neural Networks*, *184*, 107054.

Lin, E., Luo, W., Jia, W., Chen, Y., & Yang, S. (2025). Online continual learning via spiking neural networks with sleep enhanced latent replay. arXiv:2507.02901.

Liu, Y., & Wang, X.-J. (2024). Flexible gating between subspaces in a neural network model of internally guided task switching. *Nature Communications*, *15*(1), 6497.

Maass, W. (1997). Networks of spiking neurons: The third generation of neural network models. *Neural Networks*, *10*(9), 1659–1671.

Mallya, A., Davis, D., & Lazebnik, S. (2018). Piggyback: Adapting a single network to multiple tasks by learning to mask weights. In *Proceedings of the european conference on computer vision* (pp. 67–82).

Mallya, A., & Lazebnik, S. (2018). PackNet: Adding multiple tasks to a single network by iterative pruning. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7765–7773).

Mante, V., Sussillo, D., Shenoy, K. V., & Newsome, W. T. (2013). Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature*, *503*(7474), 78–84.

Maryada , Soldado-Magraner, S., Sorbaro, M., Laje, R., Buonomano, D. V., & Indiveri, G. (2025). Stable recurrent dynamics in heterogeneous neuromorphic computing systems using excitatory and inhibitory plasticity. *Nature Communications*, *16*(1), 5522.

McCloskey, M., & Cohen, N. J. (1989). Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of learning and motivation* (pp. 109–165). Elsevier (*vol. 24*).

Mendoza-Halliday, D., & Martinez-Trujillo, J. C. (2017). Neuronal population coding of perceived and memorized visual features in the lateral prefrontal cortex. *Nature Communications*, *8*(1), 15471.

Parisi, G. I., Kemker, R., Part, J. L., Kanan, C., & Wermter, S. (2019). Continual lifelong learning with neural networks: A review. *Neural Networks*, *113*, 54–71.

Rolnick, D., Ahuja, A., Schwarz, J., Lillicrap, T., & Wayne, G. (2019). Experience replay for continual learning. *Advances in Neural Information Processing Systems*, *32*.

Shaban, A., Bezugam, S. S., & Suri, M. (2021). An adaptive threshold neuron for recurrent spiking neural networks with nanodevice hardware implementation. *Nature Communications*, *12*(1), 4234.

Shen, J., Ni, W., Xu, Q., & Tang, H. (2024). Efficient spiking neural networks with sparse selective activation for continual learning. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 611–619). (*vol. 38*).

Shi, Q., Liu, F., Li, H., Li, G., Shi, L., & Zhao, R. (2025). Hybrid neural networks for continual learning inspired by corticohippocampal circuits. *Nature Communications*, *16*(1), 1272.

Skatchkovsky, N., Jang, H., & Simeone, O. (2022). Bayesian continual learning via spiking neural networks. *Frontiers in Computational Neuroscience*, *16*, 1037976.

Tiwari, A., & Chauhan, A. (2025). Local timescale gates for timescale-robust continual spiking neural networks. arXiv:2510.12843.

Upadhyay, R., Phlypo, R., Saini, R., & Liwicki, M. (2025). Meta-sparsity: Learning optimal sparse structures in multi-task networks through meta-learning. arXiv:2501.12115.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, *30*.

Wang, J., Chandra, R., & Zhang, S. (2025). Experience replay addresses loss of plasticity in continual learning. arXiv:2503.20018.

Xiao, M., Meng, Q., Zhang, Z., He, D., & Lin, Z. (2024). Hebbian learning based orthogonal projection for continual learning of spiking neural networks. arXiv:2402.11984.

Yao, H., Zhou, Y., Mahdavi, M., Li, Z. J., Socher, R., & Xiong, C. (2020). Online structured meta-learning. *Advances in Neural Information Processing Systems*, *33*, 6779–6790.

Yin, L., Yu, Y., & Wang, Q. (2026). Dynamical mechanisms of astrocyte-neuron coupling for the integration of short- and long-term memory. *Nonlinear Dynamics*, *114*(2), 74.

Yoon, J., Yang, E., Lee, J., & Hwang, S. J. (2018). Lifelong learning with dynamically expandable networks. In *International conference on learning representations*.

Zenke, F., Poole, B., & Ganguli, S. (2017). Continual learning through synaptic intelligence. In *International conference on machine learning* (pp. 3987–3995).

Zhang, A., Li, X., Gao, Y., & Niu, Y. (2021). Event-driven intrinsic plasticity for spiking convolutional neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, *33*(5), 1986–1995.

Zhang, A., Zhou, H., Li, X., & Zhu, W. (2019). Fast and robust learning in spiking feed-forward neural networks based on intrinsic plasticity mechanism. *Neurocomputing*, *365*, 102–112.

Zhang, Y., Feng, J., & Min, B. (2025). Elucidating the selection mechanisms in context-dependent computation through low-rank neural network modeling. *eLife*, *13*, RP103636.

Zheng, H., Zheng, Z., Hu, R., Xiao, B., Wu, Y., Yu, F., Liu, X., Li, G., & Deng, L. (2024). Temporal dendritic heterogeneity incorporated with spiking neural networks for learning multi-timescale dynamics. *Nature Communications*, *15*(1), 277.